Package ‘BirdPipe’

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Type Package

Title R-package for managing the workflow with the BirdPipe recording device

Version 0.1.0

Description BirdPipe is an open-source recording device for automated data collection and classification of birds, bats and other vocal animals. This R-package helps managing the BirdPipe workflow and postprocessing the data, including manual validation and model calibration.

Imports jsonlite, readxl, writexl, sonicscrewdriver, terra, tuneR, sf

License GPL-3

Encoding UTF-8

LazyData true

RoxygenNote 7.3.3

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Contents

`calibrate_classifier` *calibrate_classifier*

Description

Calibrates AI-based detections to the probability scale

Usage

```
calibrate_classifier(
  BirdPipe.data = NULL,
  BirdPipe.folder,
  audio_folder = NULL,
  n = 30,
  species = NULL,
  play.sound = TRUE,
  calibrated.classifier = NULL
)
```

Arguments

`BirdPipe.data`	A BirdPipe data object
`BirdPipe.folder`	The BirdPipe data folder
`audio_folder`	The name of the audio folder within the BirdPipe data folder
`n`	The number of expert validations to perform
`species`	The species for which calibration is to be performed
`play.sound`	Whether the sound is played for user validations
`calibrated.classifier`	Earlier output from this function, used to update calibration

Value

A named list that contains calibrated classifier and the calibration data

Examples

```
BirdPipe.folder = "BirdPipe_demo_data";
BirdPipe.data = load_data(BirdPipe.folder = BirdPipe.folder);
calibrated.classifier = calibrate_classifier(BirdPipe.data = BirdPipe.data,
BirdPipe.folder = BirdPipe.folder,species = "Eurasian Bullfinch")
```

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extract_community_matrix

extract_community_matrix

Description

Extracts a community matrix from BirdPipe data

Usage

```
extract_community_matrix(
  BirdPipe.data = BirdPipe.data,
  threshold = 0,
  name.to.use = "species_name"
)
```

Arguments

`BirdPipe.data`	A BirdPipe data object
`threshold`	Classification score threshold to be used to filter the data (influences number of detected species)
`name.to.use`	Which names to use for the species

Value

A community matrix, i.e. sampling events times species matrix

Examples

```
BirdPipe.data = load_data(BirdPipe.folder = "BirdPipe_demo_data");
extract_community_matrix(BirdPipe.data = BirdPipe.data);
```

`extract_validated_community_matrix` *extract_validated_community_matrix*

Description

Extracts a validated community matrix from BirdPipe data and manual validation file

Usage

```
extract_validated_community_matrix(
  BirdPipe.data = BirdPipe.data,
  manual.validation.file = NULL,
  n = 1,
  validated = c("Y")
)
```

Arguments

`BirdPipe.data`	A BirdPipe data object
`manual.validation.file`	Name of excel file where manual validation results are written (without the .xlsx extension)
`n`	Number of highest classifications to utilize
`validated`	Which strings are considered as validated detections

Value

A community matrix, i.e. sampling events times species matrix

Examples

```
BirdPipe.data = load_data(BirdPipe.folder = "BirdPipe_demo_data");
extract_validated_community_matrix(BirdPipe.data = BirdPipe.data, manual.validation.file = "detections_valid
```

load_data *load_data*

Description

Loads BirdPipe geojson data: recordings and species detections

Usage

```
load_data(BirdPipe.folder, threshold = 0, geojson.folder = "geojson")
```

Arguments

`BirdPipe.folder`	The name of the BirdPipe folder
`threshold`	Classification score threshold to be used to filter the data
`geojson.folder`	The name of the geojson folder within the BirdPipe data folder

Value

A named list with elements $recordings and $detections

A BirdPipe data object

Examples

```
load_data(BirdPipe.folder = "BirdPipe_demo_data");
```

play_sound *play_sound*

Description

Plays part of audio file (.flac or .wav) e.g. for manual validation purpose

Usage

```
play_sound(
  BirdPipe.folder,
  audio_folder = NULL,
  audio_filename,
  offset,
  classification.interval = 3,
  buffer = 1
)
```

Arguments

`BirdPipe.folder`	The BirdPipe data folder
`audio_folder`	The name of the audio folder within the BirdPipe data folder
`audio_filename`	The name of the audio file
`offset`	The beginning of the 3 second segment from which the vocalization is classified
`classification.interval`	Length of the classification interval in seconds
`buffer`	Number of seconds added before and after the classification interval

Examples

```
play_sound(BirdPipe.folder = "BirdPipe_demo_data", audio_filename = "field-4_2025-12-22T10_05_03.011300Z.wav
#This function call plays a vocalization of Eurasian Bullfinch from the R-package demonstration data.
#Note that Mac users may obtain a "Permission denied" error. If this happens, you may need to set the folder wher
```

`plot_detections` *plot_detections*

Description

Plots BirdPipe detections as timeline

Usage

```
plot_detections(
  BirdPipe.data,
  name.to.use = "species_name",
  species.order = "alphabetical",
  jitter = 0,
  threshold = 0
)
```

Arguments

`BirdPipe.data`	A BirdPipe data object
`name.to.use`	Which names to use for the species
`species.order`	In which order to show the species ("alphabetical" or "number_of_detections")
`jitter`	the amount of random jitter added to shows overlapping datapoints
`threshold`	Classification score threshold to be used to filter the data (influences number of detected species)

Examples

```
BirdPipe.data = load_data(BirdPipe.folder = "BirdPipe_demo_data");
plot_detections(BirdPipe.data = BirdPipe.data)
```

plot_recordings *plot_recordings*

Description

Plots BirdPipe recordings as timeline and on a map

Usage

```
plot_recordings(
  BirdPipe.data,
  show.timeline = TRUE,
  show.map = TRUE,
  background.map = NULL,
  show.number.of.species = FALSE,
  threshold = 0,
  show.sampling.event.id = FALSE,
  number.of.species.color = "white",
  sampling.event.id.color = "white",
  jitter = 0,
  spatial.buffer = 1
)
```

Arguments

`BirdPipe.data`	A BirdPipe data object
`show.timeline`	Boolean variable (TRUE/FALSE) describing whether timeline is to be plotted
`show.map`	Boolean variable (TRUE/FALSE) describing whether map is to be plotted
`background.map`	optional terra/SpatRaster object to be plotted as background map
`show.number.of.species`	Boolean variable (TRUE/FALSE) describing whether number of detected species is shown
`threshold`	Classification score threshold to be used to filter the data (influences number of detected species)
`show.sampling.event.id`	Boolean variable (TRUE/FALSE) describing whether sampling event id:s are shown
`number.of.species.color`	color for plotting number of species
`sampling.event.id.color`	color for plotting sampling event id:s
`jitter`	the amount of random jitter added to spatial locations to avoid their possible overlap (proportional to data extent)
`spatial.buffer`	size of margins added to the map (proportional to data extent)

Examples

```
BirdPipe.data = load_data(BirdPipe.folder = "BirdPipe_demo_data");
plot_recordings(BirdPipe.data = BirdPipe.data)
```

top_n_classifications *top_n_classifications*

Description

Finds the n vocalization with highest classification probabilities for each species and sampling event

Usage

```
top_n_classifications(
  BirdPipe.data,
  n = 1,
  threshold = 0.5,
  manual.validation.file = NULL
)
```

Arguments

`BirdPipe.data`	A BirdPipe data object
`n`	Number of highest classifications to extract
`threshold`	Classification threshold to be used to filter the data
`manual.validation.file`	Name of excel file where manual validation results will be written (without the .xlsx extension)

Value

A list of length n. The i:th item is a named list where the locations of i:th highest classifications are given for each species

Examples

```
BirdPipe.data = load_data(BirdPipe.folder = "BirdPipe_demo_data");
top_n_classifications(BirdPipe.data = BirdPipe.data);
```

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